GRADIENT DESCENT MANUAL CALCULATION

$y = m_1 x_1 + m_2 x_2 + b$, $m = [-1, 2]$, $b = [1, 1]$

for $x = ((1,3), (4,10))$

$y = (5, 6)$

1. $\hat{y} = mX + b$

$$\hat{y} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} (-1)(1) + (3)(2) \\ (4)(-1) + (10)(2) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 16 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$$

2. $J(m,b) = \frac{1}{2N} \|\hat{y} - y\|^2 = \frac{1}{2N} (Xm + b - y)^T (Xm + b - y)$

let error vector be $e = \hat{y} - y = \begin{bmatrix} 6-5 \\ 17-6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$ where $N = 2$

Derivative with respect to m:

$$\frac{dJ}{dm} = \frac{1}{N} X^T (mX + b - y) = \frac{1}{N} X^T e$$

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Derivative with respect to b:

$$\frac{dJ}{db} = \frac{1}{N} 1^T (mX + b - y) = \frac{1}{N} \sum e$$

3. Iterative updates ($\alpha = 0.01$)

Iteration 1 (Mutaganzwa Gift Desire)

$$e = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

~~$MSE = \frac{1}{2(2)} (1^2 + 11^2) = \frac{122}{4} = 30.5$~~

$$MSE = \frac{1}{2} (1^2 + 11^2) = \frac{1}{2} (122) = 61.0$$

$$\frac{dJ}{dm} = \frac{2}{N} X^T e^{(1)} = \frac{2}{3} \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} (1)(1) + (4)(11) \\ (3)(1) + (10)(11) \end{bmatrix}$$

$$= \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$\frac{dJ}{db} = \frac{2}{N} \sum e^{(1)} = \frac{2}{2} (1 + 11) = 12 = 12 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

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$$m_1 = m_0 - \alpha \frac{dJ}{dm} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} - 0.01 \begin{bmatrix} 45 \\ 113 \end{bmatrix} = \begin{bmatrix} -1 - 0.45 \\ 2 - 1.13 \end{bmatrix}$$

$$= \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$$

$$b_1 = b_0 - \alpha \frac{dJ}{dm} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.01 \begin{bmatrix} 12 \\ 12 \end{bmatrix} = \begin{bmatrix} 1 - 0.12 \\ 1 - 0.12 \end{bmatrix} = \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$$

Summary of findings

Iteration 1 : MSE = 61.0, $m = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$, $b = \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$

Iteration 2 : MSE = 6.8450, $m = \begin{bmatrix} -1.33 \\ 1.18 \end{bmatrix}$, $b = \begin{bmatrix} 0.93 \\ 0.93 \end{bmatrix}$

Iteration 3 : MSE = 2.72, $m = \begin{bmatrix} -1.36 \\ 1.09 \end{bmatrix}$, $b = \begin{bmatrix} 0.93 \\ 0.93 \end{bmatrix}$

Trend in values of m and b

m_1 shifts from $1.0 \rightarrow 1.45 \rightarrow 1.33 \rightarrow 1.36$

m_2 shifts from $2.0 \rightarrow 0.87 \rightarrow 1.18 \rightarrow 1.09$

Observation: The weight vector undergoes corrective adjustments in opposite directions during the initial steps to counter the high over-prediction, stabilizing

as the algorithm steps closer to the optimal fit.

- b shifts from $1.00 \rightarrow 0.88 \rightarrow 0.931 \rightarrow 0.936$

Observation: The bias term scales downward initially to lower the overall intercept of the predicted surface then finely tunes upward slightly to balance the regression hyperplane.

→ They both move in a direction that reduces the error.

This is clearly demonstrated by the rapid and continuous drop in the MSE cost function across the iterations.

The gradient descent algorithm is successfully working as intended: the computed negative gradients dictate a step vector that directly minimizes the residuals $(\hat{y} - y)$, driving the cost function down toward its global minimum.

Maths Formative 3: G15,

Given: . Initial $m = [-1, 2]$

initial $b = [1, 1]$

$\cdot x = ((1,3), (4,10))$

$\cdot y = (5, 6)$

Iteration 2:

~~Step~~ 1: Computing predictions ($\hat{y}$):

$$\hat{y} = X m^{(1)} + b^{(1)} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$$

$$= \begin{bmatrix} 1.16 \\ 2.90 \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix} = \begin{bmatrix} 2.04 \\ 3.78 \end{bmatrix}$$

~~Ste~~ 2: Computing Error vector (e)

$$e^{(2)} = \hat{y}^{(1)} - y = \begin{bmatrix} 2.04 \\ 3.78 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix}$$

3: Computing MSE cost: $MSE^{(2)} = \frac{1}{2}\left((-2.96)^2 + (2.22)^2\right)$

$$= \frac{1}{2}\left(8.7616 + 4.9284\right)$$

$$= \frac{13.69}{2} \approx \boxed{6.845}$$

4. Computing Gradients

$$\frac{\partial J}{\partial m} = \frac{2}{2} X^T e^{(2)} = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix} = \begin{bmatrix} (1)(-2.96) + (4)(-2.22) \\ (3) \quad (-2.9) + (10)(-2.2 \end{bmatrix}$$

$$= \begin{bmatrix} -11.84 \\ -31.08 \end{bmatrix}$$

$$\frac{\partial J}{\partial b} = \frac{2}{2} \sum e^{(2)} = \left(-2.96 + (-2.22)\right) = \text{~~-5.183~~} \quad \underline{-5.18}$$

5. Updating parameters

$$m^{(2)} = m^{(1)} - \alpha \frac{\partial J}{\partial m} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} - 0.01 \begin{bmatrix} -11.84 \\ -31.08 \end{bmatrix}$$

$$= \begin{bmatrix} -1.45 + 0.1184 \\ 0.87 + 0.3108 \end{bmatrix} = \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix}$$

$$b^{(2)} = b^{(1)} - \alpha \frac{\partial J}{\partial b} = \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix} - 0.01(-5.18)$$

$$= \begin{bmatrix} 0.88 + 0.0518 \\ 0.88 + 0.0518 \end{bmatrix}$$

$$= \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix}$$

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Iteration 3

1) Compute Predictions (ŷ)

$$\hat{y}^{(2)} = Xm^2 + b^2 = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.3316 \\ 1.2808 \end{bmatrix} + \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix} = \begin{bmatrix} 2.2108 \\ 4.8152 \end{bmatrix}$$

$$+ \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix} = \begin{bmatrix} 3.1426 \\ 5.7470 \end{bmatrix}$$

2) Compute Error Vector (e) -

$$e^3 = y^2 - \hat{y} = \begin{bmatrix} 3.1426 - 5 \\ 7.4134 - 6 \end{bmatrix} = \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix}$$

3) Compute MSE Cost

$$MSE^{(3)} = \frac{1}{2}\left((-1.8574)^2 + (1.4134)^2\right) =$$

$$\frac{1}{2}(3.44999 + 1.9977) = 2.7238$$

4) Compute Gradients:

- $\frac{\partial j}{\partial m} = x^T e^{(3)} = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix} = \begin{bmatrix} 3.7962 \\ 8.5618 \end{bmatrix}$

- $\frac{\partial j}{\partial b} = \sum e^3 = -1.8574 + 1.4134 = -0.444$

5) Updated Peremeters

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- $m^3 = m^2 - \alpha \frac{\partial J}{\partial m} = \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix} - 0.01 \begin{bmatrix} 3.7962 \\ 8.5618 \end{bmatrix} =$

$$= \begin{bmatrix} -7.369562 \\ 1.095182 \end{bmatrix}$$

- $b^3 = b^2 - \alpha \frac{\partial J}{\partial b} = \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix} - 0.01(-0.0000) =$

$$= \begin{bmatrix} 0.93624 \\ 0.93624 \end{bmatrix}$$